

CARIBBEAN EXAMINATIONS COUNCIL
INFORMATION TECHNOLOGY
General Proficiency

PAPER 02

2 ½ Hours

March 19, 2010

INSTRUCTIONS TO CANDIDATES

This paper consists of FOUR sections.

Answer ALL question from Section 1, and 2 should be attempted by all students. Students not doing CXC may leave at the end of this. Students doing CXC should attempt Sections 3 and 4. Use sentences and diagrams where appropriate. All answers should be made in

Standard English and with clear notations being used.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

SECTION 1: HARDWARE AND SOFTWARE

Answer ALL questions

1. Computers are made up of different components and the names of the components are often written as acronyms.

Write the meaning of EACH of the following computer acronyms and state ONE task that is performed by EACH of the components.

- a) CU b) ALU c) RAM d) ROM (8 marks)

Total 8 marks

2. State the type of optical disk that is suitable in EACH of the following situations:

- a) Best for storing videos
b) Can be reused as necessary
c) Commonly used for distributing software

(3 marks)

Total 3 marks

3. Figures A to C represents three different data capture methods.

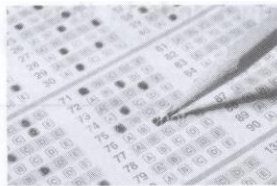


Figure A



Figure B

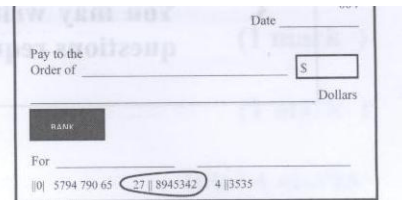


Figure C

In your answer book, copy and complete Table 1 using the list of type of data capture method below.

OMR, OCR, MICR, Voice Recognition

TABLE 1

Figure	Type of Data Capture Method
A	
B	
C	

Total 3 marks

4. a) You wish to purchase one of the devices shown in Figure 1 to 4.

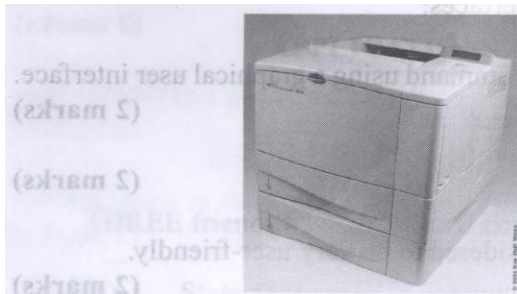


Figure 1

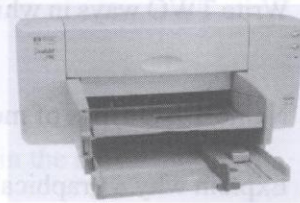


Figure 2



Figure 3

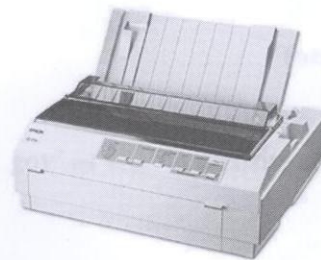


Figure 4

Table 2 lists three requirements an organization may need. In your answer booklet, copy and complete Table 2 to show which device would BEST meet EACH of the requirements in the table.

TABLE 1

Organizational Requirements	Figure
Useful for multi-form sheets	
Produces multiple copies rather quickly	
Ideal for coloured flyers	

(3 marks)

- b) Write the general name given to the devices in part (a).

(1 mark)

Total 4 marks

5. Compute the following:

- The eight-bit binary representation of the decimal number 17.
- The sign and magnitude representation for the decimal number 17.
- The results of $55 - 25$, using eight bit, two's complement representation and express your answer as an eight-bit binary number.
- What is the decimal equivalent for the binary coded decimal number 0011 0101 1000?

(4 marks)

6. Answer the following questions that relate to user interfaces:

- a) Write TWO ways in which a user can input a command using a graphical user interface. (2 marks)
- b) Name TWO types of menu interface. (2 marks)
- c) Explain why a graphical user interface is considered to be very user-friendly. (2 marks)
- d) Explain how a user can input a command using the DOS operating system. (2 marks)

Total 8 marks

SECTION 2: APPLICATIONS AND IMPLICATIONS

Answer ALL questions

7. THREE friends have purchased computers to help them in the music industry.

- a) State TWO uses of computers in the music industry. (2 marks)
- b) Name TWO devices that can connect to the computer and state what they are used for. (4 marks)

Total 6 marks

8. a) State the differences between '.edu', '.org' and '.gov' as seen in website addresses. (3 marks)

b) Write the name of a well-known search engine. (1 mark)

c) You wish to access <http://caribbeannews.com>.

- i. State where you would type <http://caribbeannews.com> to access this web site. (1 mark)
- ii. State what **http** and **www** represent in (c) above. (2 marks)

Total 7 marks

9. For EACH of the following, state whether I is (A) a private area network, (B) a public network or (C) a private network with selected access from public network users.

- | | | | |
|-------------|-------------|-------------|-----------|
| a. Intranet | b) Extranet | c) Internet | (3 marks) |
|-------------|-------------|-------------|-----------|

Total 3 marks

10. a) Describe TWO ways in which a computer can be used illegally. (4 marks)
- b) Outline ONE example of how an organization can misuse its customer's personal data. (2 marks)

Total 6 marks

11. a) Give ONE advantage and ONE disadvantage of using credit or debit cards to purchase items. (2 marks)
- b) Explain TWO potential health problems that can develop from using a computer over prolonged periods. (4 marks)
- c) Give ONE positive impact and ONE negative impact that is the result of increased use of cell phones in the workplace. (2 marks)

Total 8 marks

SECTION 3: PROGRAMMING

Answer ALL questions

12. a) Explain the difference between a '*syntax error*' and a '*logic error*'. (2 marks)
- b) Explain the difference in the execution of the TWO lines of code written below:
- i. Line 1: If MARK >50 Then Print "Good"
 - ii. Line 2: If Mark >=50 Then print "Day" (2 marks)
- c) The number 50 was the value for MARK.
- i. State the line number corresponding to the above code that would print a result. (1 mark)
 - ii. Write the result that would be printed. (1 mark)

Total 6 marks

13. Answer the questions below using the following part of a program:

```
1    value1, value2: integer;
2    Minimum: real;
3
4    value1 = 30;
5    value2 = 25.9;
6
7    IF (value1 > value2)
8    THEN read ('MinA', value2)
9    ELSE writeln ('MinB is', value2);
10
```

a) Write the line numbers that contain code to identify:

- i. input (1 mark)
- ii. declaration (2 marks)
- iii. output (1 mark)

b) The program contains FOUR errors. For EACH error write the line number and the corrected line of code. (4 marks)

c) Identify the follow:

- i) A variable (1 mark)
- ii) A conditional (1 mark)
- iii) A data type (1 mark)

d) Explain the purpose of the program segment. (3 marks)

e) State the result when EACH of the following pairs of numbers is used in the corrected program:

- i) value1 = 40, value2 = 50
- ii) value1 = 50, value2 = 40
- iii) value1 = 50, value2 = 50 (6 marks)

Total 20 marks

14. If **Box1** contains the number **16** and **Box2** contains the number **44**, state whether EACH of the following statements is **TRUE** or **FALSE**.

- a) (Box1 = 16)
- b) (Box1 < Box2)
- c) (Box1 > 16) AND (Box1 < Box2)
- d) NOT (Box2 = 44)

(4 marks)
Total 4 marks

SECTION 4: PRODUCTIVITY TOOLS

Answer ALL questions.

1. **WORD PROCESSING**

Consider the letter shown below:

Home Loan Bank
12 Main Road, Georgetown, Guyana
Tel: 229-8172; Email: accounts@hlb.com

October 1, 2009

Mr. John Smith
10 Cornell Street
Georgetown

Dear Mr. John Smith:

THIRD QUARTER LOAN PAYMENT

Please be informed that your loan payment for the third quarter of 2009 is \$12,800 and it is due on November 30, 2009¹. Please visit the loan section of the bank to discuss any concerns that you may have with respect to your loan payment, with one of our officers.

In 2010, we plan to expand the loan services we are offering to our customers. Please give us any suggestions you may want us to consider.

Yours Sincerely,

Michael Singh
Loans Manager

¹ Late payment will attract a 10% interest

- (a) Write THREE formatting features found in the letter. (3 marks)
- (b) You wish to replace all occurrences of the word 'loan' with 'Loan'.

State the feature of the word processing software you would use to achieve this.

(1 mark)

(c) What line spacing and justification have been used on the **first** paragraph of the body of the letter? (2 marks)

(d) State which of the following is used in the letter and indicate where:

- Header
- Footer
- Footnote
- Endnote

(2 marks)

(e) The secretary of the loan section wishes to merge this letter with other customer letters. Name TWO documents which are required to perform this merge. (2 marks)

(f) In the body of the letter, state TWO pieces of information that should be replaced by merge fields. (2 marks)

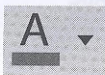
(g) State, with reasons, the number of merge fields that you should have in your letter. (2 marks)

(h) Explain how you would change the line spacing of the **second** paragraph in the body of the letter to single spacing. (2 marks)

(i) In the text 'H₂O', what formatting feature has been applied to the '2'? (1 mark)

(j) For EACH of the icons labelled (1), (2) and (3) below, state its use in a word processing program:

(1)



(2)



(3)



(3 marks)

Total 20 marks

SPREADSHEETS

Rashida and her friends decided to share their Top-40 favourite songs. Some of the songs are shown in Figure 1 below.

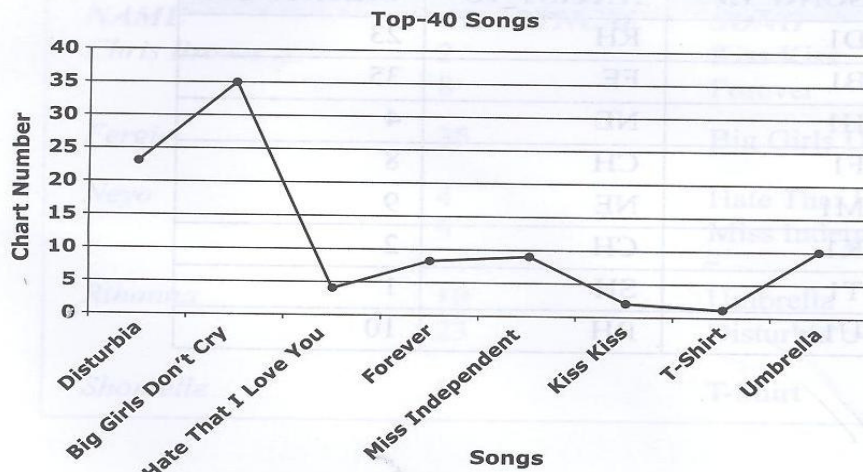
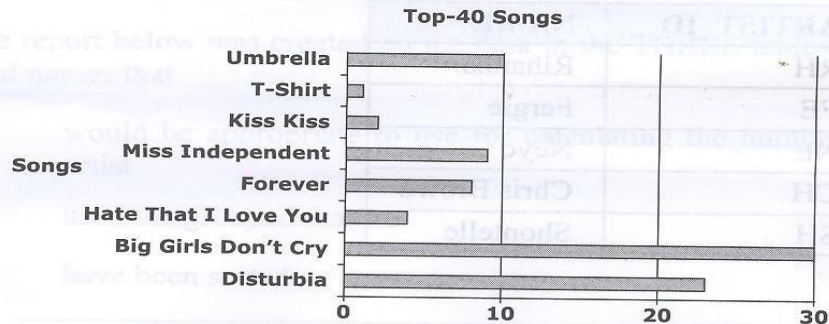
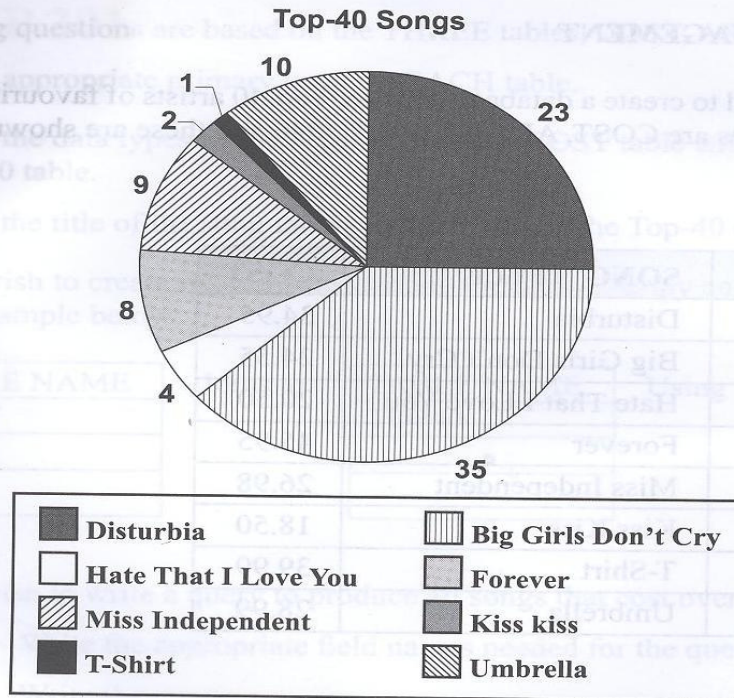
	A	B	C	D	E
1	SONG	ARTIST	Top-40	COST	OWNER
13	Disturbia	Rihanna	23	34.95	Rashida
14	Big Girls Don't Cry	Fergie	35	24.55	Rena
15	Hate That I Love You	Neyo	4	20.50	Shellaya
16	Forever	Chris Brown	8	45.95	Reyda
17	Miss Independent	Neyo	9	26.98	Alison
18	Kiss Kiss	Chris Brown	2	18.50	Kimberley
19	T-Shirt	Shontelle	1	39.99	Johnathan
20	Umbrella	Rihanna	10	28.99	Mikel

Figure 1

- (a) State the number of rows and columns shown in Figure 1. **(2 marks)**
- (b) Give the range of cells that contain the data on the songs in Figure 1. **(2 marks)**
- (c) Write functions or formulae to calculate
 - (i) the average cost of the songs **(2 marks)**
 - (ii) count the number of songs **(2 marks)**
 - (iii) the minimum cost of the songs. **(2 marks)**
- (d) State the feature that would cause Row 1 to remain visible on the spreadsheet as the user scrolls down the rows. **(1 mark)**
- (e) You decided to create a chart to show how the songs are performing on the Top-40.
 - (i) State the ranges that should be selected to create the chart. **(3 marks)**
 - (ii) THREE charts, CHART 1, CHART 2 and CHART 3, are created using the data in Figure 1 and these are shown on page 5.
 - a) For EACH chart, write the number and the corresponding chart type in your answer booklet.
 - b) Write the chart number that shows how the songs are performing. **(4 marks)**
 - (iii) State why the chart identified in (ii) b) is appropriate for that purpose. **(1 mark)**
 - (iv) Give the name of the box containing the songs in CHART 1. **(1 mark)**

Total 20 marks

CHARTS to accompany Question 2 part (e).



2. SPREADSHEETS

Rashida and her friends decided to share their Top-40 favourite songs. Some of the songs are shown in Figure 1 below.

	A	B	C	D	E
1	SONG	ARTIST	Top-40	COST	OWNER
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Figure 1

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- (c) Write functions or formulae to calculate
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 - (iii) State why the chart identified in (ii) b) is appropriate for that purpose. **(1 mark)**
 - (iv) Give the name of the box containing the songs in CHART 1. **(1 mark)**

Total 20 marks

3. DATABASE MANAGEMENT

Rashida has decided to create a database with her Top-40 artists of favourite songs. The names of the THREE tables are COST, ARTIST, and TOP40 and these are shown below.

COST Table

SONG_ID	SONG_TITLE	COST
D1	Disturbia	34.95
B1	Big Girls Don't Cry	24.55
H1	Hate That I Love You	20.50
F1	Forever	45.95
M1	Miss Independent	26.98
K1	Kiss Kiss	18.50
T1	T-Shirt	39.99
U1	Umbrella	28.99

ARTIST Table

ARTIST_ID	NAME
RH	Rihanna
FE	Fergie
NE	Neyo
CH	Chris Brown
SH	Shontelle

TOP40 Table

SONG_ID	ARTIST_ID	CHARTNUM
D1	RH	23
B1	FE	35
H1	NE	4
F1	CH	8
M1	NE	9
K1	CH	2
T1	SH	1
U1	RH	10

The following questions are based on the THREE tables, COST, ARTIST and TOP40.

- (a) Write appropriate primary keys for EACH table. (4 marks)
- (b) Write the data types for the Cost field in the COST table and the Chartnum field in the TOP40 table. (2 marks)
- (c) Write the title of the song that is number one on the Top-40 chart. (1 mark)
- (d) You wish to create relationships among the tables. Copy and complete the table using the example below:

TABLE NAME	Joins to	TABLE NAME	Using	PRIMARY KEY(S)
COST		TOP40		SONG_ID

(3 marks)

- (e) You wish to write a query to produce all songs that cost over \$40.00.
 - (i) Write the appropriate field names needed for the query. (2 marks)
 - (ii) Write the query. (2 marks)
 - (iii) Write the result of the query. (2 marks)
- (f) The report below was created on the data in the THREE tables. Write ONE or more field names that
 - (i) would be appropriate to use for calculating the number of songs for EACH artist (1 mark)
 - (ii) used the grouped feature (1 mark)
 - (iii) have been sorted. (2 marks)

Artists with Songs on Top-40 Charts

NAME	CHARTNUM	SONG
Chris Brown	2	Kiss Kiss
	8	Forever
Fergie	35	Big Girls Don't Cry
Neyo	4	Hate That I Love You
	9	Miss Independent
Rihanna	10	Umbrella
	23	Disturbia
Shontelle	1	T-Shirt

Total 20 marks

END OF TEST